STADIO SCHOOL OF INFORMATION TECHNOLOGY (SOIT)
Postgraduate Diploma in Data Science

CAP182 Capstone Project
SS2 Part A

Literature Review

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Literature Review

This review summarises four sources that inform the modelling approach for predicting account activation at STADIOEquities, a fintech investment platform with 2.3 million registered accounts. Each entry gives the Harvard reference, the modelling and preprocessing choices, the dataset, and the takeaways for the project.

Overview of the sources

Source	Setting and data	Models compared	Main result	Relevance to activation
Weterings et al. (2019)	Pension fund, 50,000 participants	Bi-LSTM with attention vs LSTM, k-NN, logistic, SVM, XGBoost	Comparable accuracy to leading models, with per-customer explanations	Predicts activation and explains which events drove it
Isen et al. (2026)	Review of 70+ studies, 2013–2025	Churn, CLV, lead scoring and segmentation models	15–24% lower CAC; up to 35% higher LTV with integrated analytics	Frames activation as a funnel outcome
Moro, Cortez and Rita (2014)	Portuguese bank, 52,944 calls, 2008–2013	Logistic regression, decision tree, neural network, SVM	NN AUC 0.794 on later data; 79% of buyers from half the calls	Time-based validation and lift-based targeting
UCI Bank Marketing (2014 release)	Same bank, 45,211 rows, 16 inputs	Benchmark dataset (no model)	Two classes; 11.7% positive; leakage and sentinel issues	Practice data for checking preprocessing choices

1. Weterings, Bromuri and van Eekelen (2019): deep attention models for activation

Title	Explaining customer activation with deep attention models'
Harvard reference	Weterings, K., Bromuri, S. and van Eekelen, M. (2019) 'Explaining customer activation with deep attention models', in *Proceedings of the 27th European Conference on Information Systems (ECIS)*, Stockholm & Uppsala, Sweden, Research Paper 151. Available at: https://aisel.aisnet.org/ecis2019_rp/151/ (Accessed: 18 October 2026).
Link	https://aisel.aisnet.org/cgi/viewcontent.cgi?article=1150&context=ecis2019_rp
Modelling	• Main model: bi-LSTM with attention and a sigmoid output, trained with binary cross-entropy. • Comparators: LSTM without attention, k-NN, logistic regression, RBF-SVM, XGBoost and a naive baseline. • Tuning: grid search with 10-fold cross-validation (32 neurons, batch size 128, 25 epochs, embedding size 60). Regularisation by dropout only.
Preprocessing	• Each participant's event history is split into prefixes, labelled positive if activation followed. This gives 135,189 sequences from 50,000 participants (37% positive). • Events are padded to a fixed length; ten demographic variables are one-hot encoded and normalised. 80/20 train–test split.
Dataset	Pension-fund data from APG, a Dutch pension executor, not publicly available. The sample is participants inactive for at least two years; activation means logging in to the personal online page after a sequence of events.
Key takeaways	• XGBoost is a strong baseline, and the attention network was not more accurate. This supports a logistic regression baseline, XGBoost as the primary model and SHAP for explanations. • Build examples from prefixes with a fixed cut-off and a 90-day outcome window, using only features from before the cut-off to avoid leakage. • Absence carries signal: the model weighted padded 'no event' slots. Include days since last login and days with no activity.

2. Isen et al. (2026): customer analytics and the sales funnel

Title	The impact of customer analytics on sales funnel conversion and customer retention in the e-commerce industry
Harvard reference	Isen, B., Moyosore, F.O., Adetola, T.P., Lawal, A.K., Amuna, M., Ntim, P.Y. and Kwarteng, R.A. (2026) 'The impact of customer analytics on sales funnel conversion and customer retention in the e-commerce industry', *Asian Journal of Economics, Business and Accounting*, 26(1), pp. 207–224. Available at: https://doi.org/10.9734/aajeba/2026/v26i1/149903.
Link	https://journalajeba.com/index.php/AJEBA/article/view/2136/4961
Modelling	• Churn: logistic regression, random forest, gradient boosting, neural networks and LSTM (reported accuracy 85–93%). • CLV: regression, Gamma-Gamma and Beta-Geometric models, neural networks (R^2 0.75–0.92). Lead scoring: logistic regression, ensemble boosting, CNN and RNN (AUC 0.82–0.95). Segmentation: K-means, hierarchical, DBSCAN and RFM clustering. • Cross-validation, hyperparameter search, bagging, boosting, stacking and threshold optimisation.
Preprocessing	• Consolidate multi-touchpoint data, impute missing values and merge fragmented accounts. • Engineer RFM, CLV, engagement and temporal features; encode categoricals; check multicollinearity (VIF, mutual information); scale; apply SMOTE for class imbalance.
Dataset	Systematic review of the literature. Primary sources include e-commerce transaction logs, SaaS usage, subscription and marketing automation data. Public datasets cited include UCI Online Retail II, Kaggle datasets and KKBOX churn data.
Key takeaways	• Judge the funnel on bottom-line outcomes (conversion, retention, LTV and LTV:CAC), not clicks or impressions. • Churn and activation are inverse problems, so churn methods such as propensity scoring and risk segmentation transfer to activation. • Ensembles and neural networks performed best in the review. Compare ensembles and consider an LSTM for sequential behaviour.

3. Moro, Cortez and Rita (2014): predicting bank telemarketing success

Title	A data-driven approach to predict the success of bank telemarketing
Harvard reference	Moro, S., Cortez, P. and Rita, P. (2014) 'A data-driven approach to predict the success of bank telemarketing', *Decision Support Systems*, 62, pp. 22–31. Available at: https://doi.org/10.1016/j.dss.2014.03.001.
Link	https://doi.org/10.1016/j.dss.2014.03.001
Modelling	• Four binary classifiers: logistic regression, decision tree (CART), neural network (one hidden layer, H = 6, averaged over 7 networks) and SVM (Gaussian kernel, γ = 2^−7.8, C = 3). • Explained with a data-based sensitivity analysis and a decision tree fitted to the network's outputs.
Preprocessing	• Inputs to the NN and SVM standardised to zero mean and unit variance. • Time-ordered split: training to June 2012, testing on the most recent 1,293 contacts. Random 2/3–1/3 train/validation split, repeated 20 times. • Domain-grouped feature selection with forward selection; rolling window of 20,000 contacts.
Dataset	52,944 contacts from May 2008 to June 2013 with 150 candidate features (reduced to 22) and a binary success/failure outcome. Only 6,557 successes (12.38%). Attributes cover clients, products, telemarketing and social and economic indicators.
Key takeaways	• Use a time-based test set: validation AUC of 0.929 fell to 0.794 on later data. • Tune only on training and validation data, touch the test set once, and test whether differences between models are significant. • Evaluate with AUC and lift, express results as business outcomes, and explain black-box models with sensitivity analysis or a surrogate tree.

4. UCI Bank Marketing dataset

Name	Bank Marketing
Harvard reference	Moro, S., Cortez, P. and Rita, P. (2014) *Bank Marketing* [Dataset]. UCI Machine Learning Repository. Available at: https://doi.org/10.24432/C5K306 (Accessed: 18 October 2026).
Link	https://archive.ics.uci.edu/dataset/222/bank+marketing
Number of instances	45,211 rows
Number of feature columns	16 input features (7 numeric, 9 categorical), plus the target column y (17 columns in total)
Number of classes	2 (yes, no)
Target range	0 and 1 after encoding
Origin	A Portuguese retail bank's telemarketing campaigns to sell long-term deposits, published on the UCI repository. The 52,944-record extract in the 2014 paper is a larger version of the same source data.
Current state (data health)	• Completeness: no blank cells are reported, but 'unknown' codes appear, most in poutcome (about 82% of rows) and contact (about 29%). • Sentinel value: pdays uses −1 for 'never contacted' (about 82% of rows), which will distort models if treated as a number. • Class balance: about 11.7% 'yes', so accuracy alone is misleading. Leakage risk: duration is known only after the call and should be excluded.